SECTION – 14

TECHNICAL SPECIFICATION
FOR
SUPPLY OF GALVANISED 220KV AND 132KV TRANSMISSION LINE TOWERS, EXTENSIONS AND TOWER ACCESSORIES

1.1 SCOPE:

This part of the specification provides for procurement of steel, proto fabrication, assembly into tower for inspection, mass fabrication, galvanizing and delivery of transmission line towers of 220 kV and 132kV comprising super-structures, Templates, stubs and cleats, tower extensions, and tower accessories such as Hangers, U-bolts, D-shackles, Extension Links etc., based on the general assembly drawings to be supplied by APTRANSCO.

1.2 Quoted prices shall include cost of steel and Zinc. It is not the intent to specify completely herein all the details of design and fabrication of Towers. However, the Towers shall conform in all respects to high standards of Engineering, Design and Workmanship and shall be capable of performing in continuous commercial operation. The offered towers shall be of complete shape in all aspects with all the parts & accessories for an effective and trouble free performance. All such things shall be deemed to be within the scope of the Bidder's supply irrespective of whether those are specifically brought out in this Specification and/or commercial order or not.

2.0 <u>STANDARDS:</u>

Except as modified in this specification, the materials covered under "scope" vide clauses 1.1 to 1.2, shall conform to the latest revision with amendments thereof of the following Bureau of Indian Standards and other International Standards.

Sl. No.	Bureau of Indian Standards (BIS)	Title
1.	IS:209	Specification for Zinc
2.	IS:2062	Structural steel (standard quality)
3.	IS:432	Mild steel and medium tensile bars and hard drawn steel wire for concrete reinforcement
4.	IS:802	Code of practice for use of structural steel in overhead transmission line Part I : Loads and permissible Stresses Part II : Fabrication, Galvanizing, Inspection & Packing
5.	IS:1367	Supply conditions for threaded fasteners (First Revised)
6.	IS:2016	Plain washers
7.	IS:2551	Danger Notice Plates

Sl. No.	Bureau of Indian Standards (BIS)	Title
8.	IS:2629	Recommended practice for hot dip galvanizing of Iron and Steel
9.	IS:2633	Method of testing uniformity of casting of zinc coated articles
10.	IS:3063	Single coil rectangular section spring washers for bolt nuts, screws
11.	IS:5358	Hot dip Galvanised coatings on fasteners
12.	IS:6610	Specification for heavy washers for steel structures
13.	IS:12427	Hexagonal bolts for steel structures
14.	IS:6745	Methods of determination of weight of zinc coating of zinc coated iron and steel articles
15.	IS:5613	Code of practice for Design Part I & II Installation & Maintenance of Section of Overhead Power Line
16.	IS:8500	Structural Steel Micro Alloyed (MEDIUM AND HIGH STRENGTH QUALITY)
17.		Indian Electricity Rules 1956
18.		Publication for Regulation for Electrical Crossing of Railway tracks
19.	IS: 4759	Hot dip zinc coating on structural steel and other allied products, plain and heavy washers, spring washers Electro galvanization.

3.0 PRINCIPAL PARAMETERS:

3.1 The steel used for fabrication of tower parts of normal type, tower extensions, stubs, stub templates etc., shall be of mild steel of tested quality as per latest versions of IS-2062.

3.2 Steel sections used in the design of towers shall be in conformity to IS-802. Dimensional and mass tolerances of the various sections shall conform to the appropriate values stipulated in IS-1852. No individual member shall be longer than 6000 mm.

4.0 The following sizes of steel sections are used in the design of towers with tolerances as per IS-1852-1985.

Equal Angles

i)	200x200x20/16/24/25	ix)	75x75x8/6/5
ii)	150x150x20/16/12	x)	70x70x5
iii)	130x130x12/10	xi)	65x65x6/5
iv)	110x110x10/8	xii)	60x60x5/4
v)	100x100x10/8/6	xiii)	55x55x5/4
vi)	90x90x8/6	xiv)	50x50x6/5/4
vii)	80x80x8/6	xv)	45x45x5/4
viii)	250x250x25/24/20/16		

Unequal Angles

45x30x4/5

Plates

i)	25mm thick	vi)	8 mm thick
ii)	20mm thick	vii)	6 mm thick
iii)	16mm thick	viii)	5 mm thick
iv)	12mm thick	ix)	4 mm thick
v)	10mm thick	x)	2 mm thick

5.0 GENERAL TECHNICAL REQUIREMENTS

5.1 220 KV and 132kV Transmission Line TOWERS:

Consists of various type of tower as indicated in section – 17 of Line works (Appendix-1A). The Bidder has to inspect the same at site and prepare detailed and shop drawing and proceed for fabrication.

5.1.1 The towers shall be of self-supporting type, built up of lattice steel sections or members.

5.1.2 The towers shall be fully Galvanized structures built up of structural mild steel sections. All members shall be connected with bolts, nuts and spring washers.

5.1.3 Stubs and Superstructures and extensions:

i) The stubs shall mean a set of four stub angles galvanized from top to 300mm below ground level and rest black and shall include cleats, the black portion of the stub being cast in foundation footings.

ii) Superstructure shall mean the Galvanized tower assembly above the stubs which includes structural members like angle sections, cross arms, Earth wire peak, gusset plates and pack washers. The Bidder shall arrange steel and zinc required for manufacturing these items.

Extensions shall mean the galvanized tower assembly above the stubs, which includes structural members for maintaining adequate ground clearance.

iv) Supply of hangers for attaching suspension strings, 'U' bolts for attaching ground wire suspension assemblies are included in the scope of supply since these are required for completing the towers in all respects.

v) The following provisions shall apply in connection with the procurement of steel and zinc by the Bidder.

a) The 200x200x20 section if any to be used for leg members shall be of HT steel of tested quality IS: 8500 or any revision thereof.

b) The Bidder shall take into account the fabrication wastage while quoting the rates. The Purchaser will not accept any liability in connection with the wastage of steel during fabrication or otherwise.

c) Structural drawings will be made available to the successful Bidder only and they must take a copy and return the original supplied to them.

d) Substitutions, if any, of steel sections of the tower parts by higher sizes, due to non-availability or otherwise, shall be to the Supplier's account. The Purchaser will not accept any liability on this account.

e) The steel shall be procured exclusively from the main steel producers. However, sections not rolled by main producers, can be procured from re-rollers provided:

- Re-rolling of structural steel sections is done from billets/ingots of tested quality confirming to latest revision of IS-2062.
- Re-rolled sections are duly tested as per relevant standard.

f) The zinc used for galvanizing fabricated material shall be as per latest revision of IS-209 and of High Grade Electrolytic Zinc of 99.95% purity.

5.1.4 Extensions:

The towers are suitable for adding 3 M, 6 M, 9 M and 12 M extensions for maintaining adequate ground clearances without reducing the specified factor of safety in any manner.

5.2 Tower Accessories:

5.2.1 Insulator strings and Earthwire clamp attachments

a) For attachment of suspension insulator strings, a swinging hanger on the tower shall be supplied as per the drawing (to be furnished by the Purchaser) so as to obtain requisite clearance under extreme swinging conditions and free swinging of the string. The hanger shall be fabricated (forged) to withstand an ultimate tensile strength of 7,000 Kgs for 132 kV towers and 14,000kgs for 220 kV towers. In case of Towers supporting twin Moose conductors, the hanger shall with stand 16,500 kgs. The hanger set shall be hot dip galvanized and shall comprise of the forged hanger rods, bolt etc., as detailed in the drawing.

b) For earth wire at suspension towers suitable 'U' Bolts shall be supplied as per the drawing (to be furnished by the Purchaser) to accommodate the hook of the Earthwire suspension clamps. The 'U' bolt shall be hot dip galvanized and shall comprise of a 'U' bolt with nuts and spring washers as detailed in the drawing.

c) For the attachment of tension insulator strings a tension D-shackle on the angle tower shall be supplied as per the drawing (to be furnished by the Purchaser) and to obtain requisite clearance at bottom and middle cross arms an extension links is required. The D-shackle shall be fabricated (forged) to withstand an ultimate tensile strength higher than the details mentioned in 5.2.1(a) above. The D-shackle along with extension link set shall be hot dip galvanized.

5.3 Details of Tower Fabrication Workmanship

5.3.1 The tower structures shall be accurately fabricated to bolt together easily at site without any strain on the bolts.

5.3.2 The diameter of the holes shall be 17.5 mm i.e., equal to the diameter of 16 mm bolt plus 1.5 mm.

5.3.3 All steel sections before any work is done on them, shall be carefully leveled, straightened and made true to detailed drawings by methods which shall not injure the materials so that when assembled, the different matching surfaces are in close contact throughout. No rough edges shall be permitted anywhere in the structure.

5.3.4 Drilling and Punching

5.3.4.1 Before any cutting work is started, all steel sections shall be carefully straightened and trued by pressure and not by hammering. They shall again be trued after being punched and drilled.

5.3.4.2 Holes for bolts shall be drilled or punched with a jig but drilled holes are preferred. The following maximum tolerance of accuracy of punched holes is permissible.

a) Holes must be perfectly circular and no tolerance in this respect is permissible.

b) The maximum allowable difference in diameter of the holes on the two sides of plates or angle is 0.8 mm ie., the allowable taper in punched holes should not exceed 0.8 mm in diameter.

c) Holes must be square with the plates or angles and have their walls parallel.

5.3.4.3 All burrs left by drills or punches shall be removed completely. When the tower members are in assembled position, the holes shall be truly opposite to each other. Drilling or ramming to enlarge defective holes is not permitted.

5.3.5 Erection Mark:

Standard Instructions of APTransco shall be followed regarding Erection Mark.

5.3.6 Abbreviation of suppliers name shall be separately marked on each tower member with marking dies of 16mm size. Approval of Chief Engineer/Construction for abbreviation shall be taken before manufacture of proto type & taking up the mass fabrication of towers.

5.4 Galvanising:

The superstructure of all towers and stubs upto 300mm below ground level shall be fully galvanized. Galvanizing shall be in conformity with IS: 2629, IS: 4759 and shall be done after all fabrication work has been completed. Threads of bolts and nuts in respect of hanger and U bolt after Galvanising shall have a neat fit and shall be such that they can be turned with fingers throughout the length of the threads of bolts and they shall be capable of developing the full strength of the bolts. Galvanising for U bolts shall conform to IS: 5358.

For transmission lines, substation structures and associated equipment located in coastal regions within 25 km from the seacoast or in areas identified as having severe pollution levels, The fabricated tower parts and stubs shall have a minimum overall Zinc coating of 900gms/sq.m of surface except for plates and sections below 5 mm which shall have a minimum overall Zinc coating of 610 gm/sq.m of surface area. The average Zinc coating for all sections and plates 5mm and above shall be maintained as 127 microns and that for plates and sections below 5 mm shall be maintained as 87 microns.

The zinc coating shall be adherent, reasonably uniform, smooth, continuous and free from imperfections such as black/ bare spots, ash rust strains, bulky white deposits/ wet storage strains and blisters.

5.5 Quantities and Weights:

5.5.1 The quantities and types of towers and steel sections stated in Price Schedule and the Annexure are only provisional. Final quantities and types of towers and steel sections will be informed by the Purchaser to the successful Bidder on completion of detailed surveys on the lines for which these towers are intended, while placing the detailed order.

Further, number plates, danger boards, circuit plates for both circuits, and phase plates will be compulsory on all towers across the transmission line. Additionally, anti-climbing devices will be installed at crucial locations such as NH crossings, railway crossings, power line crossings, and other important areas as identified by APTransco.

5.6 Bolt Arrangement:

The minimum bolt spacing and rolled edge distance and sheared edge distances of sections from the centers of the bolt holes to be maintained are given below.

Dia of bolts (mm)	Hole dia spacing (mm)	Min. bolt distance (mm)	Min.rolled edge distance(mm)	Min.sheared edge distance (mm)
16	17.5	40	22	25

5.7 Allowable Stress:

Structural steel angle sections manufactured according to the latest IS: 802 (Part V & VI) and tested according to the latest edition of IS: 2062 or any revision thereof and having yield strength of not less than 255 N/mm. Sq. shall be used in the fabrication of tower members.

5.8 **Packing & forwarding**

The packing shall be done in such a way that the towers received at the site stores at any instance are in full shape.

Angle sections shall be wire bundled.

Cleat angles, gusset plates, and hangers and similar other loose items shall be netted and bolted together in multiples or securely wired together through holes.

U Bolts and pack washers shall be packed in double gunny bags, accurately tagged, in accordance with the contents.

The packing shall be properly done to avoid losses/damages during transit. Each bundle or package shall be appropriately marked.

6.0 Proto Inspection

6.1 Proto assembly of each type of tower, extension, and stub setting template shall be done by the supplier and offered for inspection to the Purchaser before taking up mass fabrication of the structures required. Waiving the Proto Assemblies inspection, for various reasons including urgency is purely at the discretion of APTRANSCO.

Two sets of shop drawings/sketches for stub, superstructure, and all extensions upto 12M for all types of towers shall be furnished for verification before inspection of proto type. Six sets of copies of structural drawings and shop drawings/sketches, Bill of materials for each type of tower/structure shall be furnished after inspection of proto type. Apart from this, by the end of the approvals of all the drawings (structural, shop & BOM), 2 sets of all the drawings and Bill of Materials in CDs are to be provided to APTRANSCO. While dispatching the Towers at the given stores, copies of Bill of materials of all respective Tower parts shall be handed over to the consignee for verification.

6.2 Galvanization of the members of the tower shall withstand tests as per latest edition of IS:2633.

7.0 INSPECTION

7.1 The successful Bidder shall keep the Purchaser informed well in advance of the commencement of manufacture, progress of manufacture thereof and fabrication of various tower parts at various stages, so that arrangements could be made for inspection by the Purchaser. Towers shall be offered for inspection in such a way that they are in full shape.

7.2 The acceptance of any batch of items shall in no way relieve the Supplier of any of his responsibilities for meeting all the requirements and intent of this specification and shall not prevent subsequent rejection of any item of that batch later found defective.

7.3 The Purchaser or his authorized representatives shall have free access at all reasonable times to inspect those parts of the Supplier's works connected with the fabrication of the materials covered under the contract for satisfying themselves provisions of this specification.

7.4 Unless specified otherwise, inspection shall be made at the place of manufacture prior to despatch and shall be conducted so as not to interfere unnecessarily with the operation of the work.

7.5 Should any member of the structure be found not to comply with the approved drawing, it shall be liable for rejection. No member once rejected shall be resubmitted for inspection, except in cases where the Purchaser or his authorized representative considers that the defects can be rectified.

7.6 Defects, which occur during fabrication, shall be made good with the consent of, and according to the procedure to be laid down by the Purchaser.

7.7 All gauges and templates necessary to satisfy the Purchaser for conducting tests shall be made available at the test site by the Supplier.

7.8 The correct grade and quality of steel shall be used by the Supplier. To ascertain the quality of steel, the Purchaser may at his discretion get the material tested at an approved Laboratory.

7.9 The Supplier shall give 15 days advance intimation to enable the Purchaser to depute his representative for witnessing acceptance and routine tests. The supplier shall offer the materials for inspection in such a way that the towers reach the stores in full shape with all extensions, accessories and stubs & cleats, stub setting templates.

8.0 DOCUMENTATION

Approval of drawing/work by Purchaser shall not relieve the supplier of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices. The material shall conform in all respects to high standards of engineering, design, workmanship and latest revisions of relevant standards at the time of ordering and purchaser shall have the power to reject any work or material which in his judgment is not in full accordance therewith.

GENERAL TECHNICAL PARTICULARS FOR GALVANISED TRANSMISSION LINE TOWERS & ACCESSORIES

(to be filled and furnished for each type of Tower along with the Technical Bid)

132KV / 220 kV Towers:

1.1 Name and address of:

a) Tower fabricator
b) Galvanizer

1.2 Fabricated Tower – Guaranteed Tower Weight (Kg.)

1.3 **Sags & Tension**

		Conductor		Ground wire	
		Tension (In kg)	Sag (In m)	Tension (In kg)	Sag (In m)
a)	32^0 C (Every day temp) and Nil wind				
b)	32^0 C and Full wind				
c)	32^0 C and 75% of Full wind				
d)	0^0 C (minimum temp) and 36% of Full wind				

1.4 Standards to which conform

a) Mild steel
b) Zinc
c) U-Bolt
d) Hanger
e) Step bolts
f) Tower accessories
(List the accessories).

1.5 Minimum guaranteed yield stress;

(Kg / mm^2) corresponding angle sections – Mild steel

1.6 Maximum allowable stress (Kg / mm^2):

a) Tension Members (net area) - MS
b) Compression Members (gross area) - MS

1.7 Stress on area of class 4.6 & 5.6

	Class 4.6	Class 5.6
a) Shear stress on gross area (Kg / cm^2)		
b) Bearing stress on gross area (Kg / cm^2)		
c) Bearing stress on gross area (Kg / cm^2)		
d) Tension (Kg / cm^2)		

1.8 Size and grade of bolts and corresponding limiting stress for steel section MS:

1.9 Step Bolt:

a) Size dia (mm)
b) Overall length (mm)
c) Length of shank / Unthreaded portion bolt (mm)
d) Vertical load with stand capacity at 115 mm distance.

1.10 Minimum clearance for crossing:

a) Overhead power lines
b) Overhead telecom lines
c) Railway track
d) River crossing - Non navigable
Navigable

1.11 Galvanization:

a) Method

Weight of zinc (g / M^2)
- Steel sections
- Bolts and nuts
- Spring washers
- Pack washers

b) No. of dips of duration
1 Minute
½ Minute

c) Quality of zinc & standard to which conforming

1.12 **Fabrication tolerance and standard to which conforming:**

a) Camber (straightness)
b) Allowable taper in punched holes
c) Overall length of member
d) Consecutive holes
e) First hold and last hold
f) Specified hole diameter punched / drilled
g) Corner cut, notches, flange cut

1.13 Method of bending and bend angle for angle sections and plates:

a) Angle section upto 75 x 6 mm
Above 75 x 6 mm and upto 100 x 8 mm
Above 100 x 8 mm

b) Plates
Thickness upto 12 mm
Thickness greater than 12 mm

1.14 Punching and drilling:

Maximum thickness for
Punching
Drilling

1.15 Edge security for rolled and sheared section and bolt gauges for size of bolts used in fabricated tower:

1.16 Tower accessories:

a) **Bird Guards:**
- Governing standards
- Dimensioned drawing enclosed: Yes / No
- Galvanization method and applicable standard weight of zinc (gm / M^2) / No. of dips of duration one minute and half minute.

b) **Anti climbing device:**
- Governing standards
- Dimensioned drawing enclosed: Yes / No
- Galvanization method and applicable standard weight of zinc (gm / M^2) / No. of dips of duration one minute and half minute.

    i) Barbed wire
    ii) Bolts & Nuts
    iii) Spring washers
    iv) Structural

c) **Number plate / phase plate / name plate / danger plate:**
- Governing standards
- Materials & its thickness
- Details of paint (Base coat, finish coat)
- Dimensioned drawing enclosed: Yes / No

d) **Earthing Device:**

i) Pipe Earth:

- Governing standards
- Galvanization method and applicable standard weight of zinc (gm / M^2) / No. of dips of duration one minute and half minute.
- Dimensioned drawing enclosed: Yes / No

ii) Counterpoise Earth:

- Governing standards
- Galvanization method and applicable standard weight of zinc (gm / M^2) / No. of dips of duration one minute and half minute.
- Details of ground wire and governing standard to be used as counterpoise.

1.17 Details of packing

a) Structural Members
b) Tower Accessories

1.18 Aviation signaling Devices / Visual Aids:

- Governing standard
- Bill of material (Line markers / structure marking / medium flashing light / low intensity red.
- Stationery light (Name the components, material and indicate the quantity against each).
- Candle power of low intensity Red Stationary Light.
- Source of power for night marking.
- Dimensioned drawing of various components enclosed: Yes / No.
- Details of structure marking (color code, base treatment, type of paint etc.,).

SIGNATURE OF THE BIDDER
WITH DATE & SEAL